

# Product Design Document

*Cross-Platform Ad-Free & Voice-Activated Chanting Counter Ecosystem  
Targeting Android Mobile, Tablet, and Google TV*

## 1. Executive Summary

This document specifies the architecture, core engine design, user experiences, and data models for a premium, cross-platform digital spiritual ecosystem. The app provides a completely ad-free experience, running local machine learning modules to enable real-time, hands-free voice tracking of repetitive chants or prayers. It maintains uniform synchronization across individual smartphones, tablets, and smart TVs.

## 2. Technical Architecture & System Topography

### 2.1 On-Device AI Voice Recognition Model

- **Local VAD (Voice Activity Detection):** Employs a compressed, local footprint model optimized via TensorFlow Lite or Android's Neural Networks API (NNAPI) to perform real-time speech segregation without outbound network telemetry.
- **Acoustic Calibration:** Supports low-decibel whispered syllables and rapid vocal cadences up to 240 CPM (chants per minute).
- **Dynamic Noise Suppression:** Utilizes low-latency spectral subtraction layers to filter environment noises like home fans or background music.

### 2.2 Data Management & State Machine

- **State Integrity:** Local calculations leverage SQLite via Room Database architecture to store microsecond transaction logs.
- **Sovereign Cloud Sync:** Relies directly on Application Data Folders within user-owned Google Drive accounts using OAuth2 authentication mechanisms.

### 3. Cross-Platform Interface Frameworks

| Platform Form Factor      | Primary UI Concept                                       | Key Interaction Focus                                                          |
|---------------------------|----------------------------------------------------------|--------------------------------------------------------------------------------|
| Android Mobile Phone      | Low-friction pocket design with minimal visual glare     | Background background mic tracking, system overlay, hardware button overrides. |
| Android Tablet / Foldable | Detailed multi-panel dashboard with concurrent utilities | Split-screen digital scripture prompts and graphical metric dashboards.        |
| Google TV / Android TV    | Immersive layout visible up to 10 feet away              | Minimalist ambient visualizations and simplified directional remote mapping.   |

### 4. Immersive Spiritual Frameworks

- **Zero Distraction Policy:** Absolute code-level exclusion of software advertisement SDKs, analytics tracking pixels, and non-essential telemetry hooks.
- **Feedback Engines:** Employs physical vibration indicators (haptic pulses at intervals of 1 and unique rhythmic dual-pulses at 108 counts) alongside crystal-clear studio acoustic signals (temple bells, singing bowls).